

Course Syllabus – Fall 2026

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1. Logistics

Class: BMME 575.001

Location (Hybrid): MoWeFr 12:20PM-1:10PM, Marsico Hall - Rm 6004

Zoom link: <https://zoom.us/j/92793761804>

Class Slack Channel:

https://join.slack.com/t/bmme575-fall2026/shared_invite/zt-46hfjd3o-SEf0z01CxWo3IoaX~EuV9g

Instructor Office hours: Fridays 1:30 PM – 3:00 PM

(With appointment ONLY) **Zoom link:** <https://zoom.us/j/91421416071>

Appointments available at: <https://calendly.com/azarang1993/bmme-575-office-hours>

Course TA: Qianqian Wang (qqw@email.unc.edu)

TA Office hours: Fridays 3:00 PM – 5:00 PM

(With appointment ONLY) **Zoom link:** <https://zoom.us/j/98946096863>

Appointments available at: <https://calendly.com/qqw-ad/bmme-575-ta-office-hours>

2. Learning Objectives

By the end of the course students will be able to:

- Explore, visualize, and interpret datasets using the matplotlib and pandas Python libraries
- Identify and use properly machine learning terms and concepts: supervised, semi-supervised and unsupervised learning, variance-bias tradeoff, underfitting and overfitting, batch and online learning, testing and validating, classification and regression
- Formalize a problem in machine learning terms, solve with the appropriate method in the scikit and keras learning packages, and visualize solution metrics

- Assimilate principles and limitations of the following machine learning techniques: classification, regularized linear regression, dimensionality reduction, kernel methods, neural networks, and deep learning
- Set up a pipeline for image classification and image denoising.

3. Course Description

	Date	Lecture Content
1	8/17/2026 – Mo	Syllabus Review and Successful Stories of Deep Learning Models
2	8/19/2026 – We	Project Presentation
3	8/21/2026 – Fr	Anaconda Installation/Environment Setup/Getting Started with Python – Part 1
4	8/24/2026 – Mo	Anaconda Installation/Environment Setup/Getting Started with Python – Part 2
5	8/26/2026 – We	GitHub Introduction and Overview of GPU Resources
6	8/28/2026 – Fr	Jupyter Notebook/Pandas/Data Visualization – Part 1
7	8/31/2026 – Mo	Jupyter Notebook/Pandas/Data Visualization – Part 2
8	9/2/2026 – We	OpenCV and Matplotlib Libraries
9	9/4/2026 – Fr	Linear Algebra
	9/7/2026 – Mo	No Class (Labor Day Holiday)
10	9/9/2026 – We	Linear Algebra – Practical Examples in Jupyter Notebook
11	9/11/2026 – Fr	Calculus
12	9/14/2026 – Mo	Probability and Information Theory
13	9/16/2026 – We	PCA and Dimension Reduction followed by Practical Examples in Jupyter Notebook – Part 1
14	9/18/2026 – Fr	PCA and Dimension Reduction followed by Practical Examples in Jupyter Notebook – Part 2
	9/21/2026 – Mo	No Class (Well-being Day)
15	9/23/2026 – We	Introduction to Deep Learning – Part 1
16	9/25/2026 – Fr	Introduction to Deep Learning – Part 2
17	9/28/2026 – Mo	Introduction to Deep Learning – Part 3
18	9/30/2026 – We	Introduction to Deep Learning – Part 4
19	10/2/2026 – Fr	Design and Training Setups, Challenges in Deep Learning – Part 1
20	10/5/2026 – Mo	Design and Training Setups, Challenges in Deep Learning – Part 2
21	10/7/2026 – We	Design and Training Setups, Challenges in Deep Learning – Part 3
22	10/9/2026 – Fr	Design and Training Setups, Challenges in Deep Learning – Part 4
23	10/12/2026 – Mo	Longleaf Cluster Setup (MobaXterm, Globus, etc.) – Part 1
24	10/14/2026 – We	Longleaf Cluster Setup (MobaXterm, Globus, etc.) – Part 2
	10/16/2026 – Fr	No Class – Fall Break
25	10/19/2026 – Mo	Classification and Regression Tasks
26	10/21/2026 – We	Classification and Regression Tasks - Practical Examples in Jupyter Notebook
27	10/23/2026 – Fr	Convolutional Neural Networks – Part 1
28	10/26/2026 – Mo	Convolutional Neural Networks – Part 2
29	10/28/2026 – We	Convolutional Neural Networks – Practical Example in Jupyter Notebook

30	10/30/2026 – Fr	Small Dataset Challenges (1): Cross Validation and its variants
31	11/2/2026 – Mo	Small Dataset Challenges (2): Image Augmentation
32	11/4/2026 – We	Transfer Learning: Challenges and Practical Examples in Jupyter Notebook
33	11/6/2026 – Fr	Unsupervised Learning: Autoencoders and Generative Models
34	11/9/2026 – Mo	Autoencoders and their variants – Practical Example in Jupyter Notebook
35	11/11/2026 – We	Explainable Deep Learning Models
36	11/13/2026 – Fr	Semi- and self-supervised Learning/Foundation Models
37	11/16/2026 – Mo	Deep Learning Models on Edge Devices
38	11/18/2026 – We	Project Delivery
39	11/20/2026 – Fr	Project Delivery
40	11/23/2026 – Mo	Project Delivery
	11/25/2026 – We	No Class (Thanksgiving Recess)
	11/27/2026 – Fr	No Class (Thanksgiving Recess)
41	11/30/2026 – Mo	Project Delivery
42	12/2/2026 – We	Project Delivery
43	12/4/2026 – Fr	Project Delivery

4. Grade and Evaluations

The final grade will be calculated as follows

20% classroom quizzes

20% homework assignments

60% project milestones, final delivery and presentation

The grade for graduate students will be assigned as

High pass: > 80%

Pass: >= 60%

Fail: < 60%

The grade for undergraduate students will be assigned as

A: >93.0

A-: 90-92.9

B+: 87.0-89.9

B: 83.0-86.9

B-: 80.0-82.9

C+: 77.0-79.9

C: 73.0-76.9

C-: 70.0-72.9

D+: 67.0-69.9

D: 60.0-66.9

F: <60.0

Undergraduate and graduate students will work individually on homework.

Rubric

- 100%: The solution is clear and correct.
- 80%: The solution is clear but contains a few mistakes, but they are mostly arithmetic or of little significance.
- 60%: The solution hits on the main points but has at least one logical gap.
- 40%: The solution is significantly unclear or contains several major gaps, but parts of it are salvageable.
- 20%: The solution is just plain wrong or so unclear it cannot be followed.
- 0%: No attempt is made at solving the problem.

5. Course Resources

ACCESSIBILITY RESOURCES

The University of North Carolina at Chapel Hill facilitates the implementation of reasonable accommodations, including resources and services, for students with disabilities, chronic medical conditions, a temporary disability or pregnancy complications resulting in barriers to fully accessing University courses, programs and activities.

Accommodations are determined through the Office of Accessibility Resources and Service (ARS) for individuals with documented qualifying disabilities in accordance with applicable state and federal laws. See the ARS Website for contact information: <https://ars.unc.edu> or email ars@unc.edu.

University Policy

No right or privilege exists that permits a student to be absent from any class meetings, except for these University Approved Absences:

- Authorized University activities
- 1. Disability/religious observance/pregnancy, as required by law and approved by Accessibility Resources and Service and/or the Equal Opportunity and Compliance Office (EOC)
- 2. Significant health condition and/or personal/family emergency as approved by the Office of the Dean of Students, Gender Violence Service Coordinators, and/or the Equal Opportunity and Compliance Office (EOC).

Class policy

Instructors may work with students to meet attendance needs that do not fall within University approved absences. For situations when an absence is not University approved (e.g., a job interview or club activity), instructors determine their own approach to missed classes and make-up assessments and assignments. Please provide your approach on the course syllabus.

Please communicate with me early about potential absences. Please be aware that you are bound by the Honor Code when making a request for a University approved absence.

(source: <http://catalog.unc.edu/policies-procedures/attendance-grading-examination/>)

Honor Code statement

1. I expect all students to follow the guidelines of the UNC honor code. In particular, students are expected to refrain from “lying, cheating, or stealing” in the academic context. You can read more about the honor code at honor.unc.edu. In any course, including mine, what constitutes cheating can change from one activity to another. For example, collaboration may be encouraged for an assignment but qualify as cheating during an exam. Please see my guidelines for each activity, and if you are unsure, please ask me to clarify. In remote classes, there may be many temptations for using online exchange sites, such as Chegg. Note that these sites provide names of students who have used their materials, and they routinely cooperate with institutions around academic integrity issues. Please don’t get caught up with honor code issues just because it appears to be simple and untraceable. It is not!
2. All students are expected to follow the guidelines of the UNC honor code. In particular, students are expected to refrain from “lying, cheating, or stealing” in the academic context. If you are unsure about which actions violate that honor code, please see me or consult honor.unc.edu.
3. Students are bound by the Honor Code in taking exams and in written work. The Honor Code of the University is in effect at all times, and the submission of work signifies understanding and acceptance of those requirements. Plagiarism will not be tolerated. Please consult with me if you have any questions about the Honor Code.
4. The University of North Carolina at Chapel Hill has had a student-administered honor system and judicial system for over 100 years. The system is the responsibility of students and is regulated and governed by them, but faculty share the responsibility. If you have questions about your responsibility under the honor code, please bring them to your instructor or consult with the office of the Dean of Students or the Instrument of Student Judicial Governance. This document, adopted by the Chancellor, the Faculty Council, and the Student Congress, contains all policies and procedures pertaining to the student honor system. Your full participation and observance of the honor code is expected (honor.unc.edu).
5. As a condition of joining the Carolina community, Carolina students pledge “not to lie, cheat, or steal” and to hold themselves, as members of the Carolina community, to a high

standard of academic and non-academic conduct while both on and off Carolina's campus. This commitment to academic integrity, ethical behavior, personal responsibility and civil discourse exemplifies the "Carolina Way," and this commitment is codified in both the University's Honor Code and in other University student conduct-related policies.

Acceptable use policy

By enrolling as a student in this course, you agree to abide by the University of North Carolina at Chapel Hill policies related to the acceptable use of IT systems and services. You may be asked to participate in online discussions or other online activities that may include personal information about you or other students in the course. The rights and protection of other participants are protected under the UNC-Chapel Hill Information Technology Acceptable Use Policy, which covers topics related to using digital resources, such as privacy, confidentiality, and intellectual property.

Consult the University website "Safe Computing at UNC" for information about the data security policies, updates, and tips on keeping your identity, information, and devices safe.

Data security and privacy

Sakai's Discussion Forum, Assignments, DropBox, Gradebook, and Tests & Quizzes tools are designed to share FERPA-protected information privately between instructors and individual students.

Syllabus changes

The professor reserves the right to make changes to the syllabus, including project due dates and test dates. These changes will be announced as early as possible.

Grade appeal process

If you feel you have been awarded an incorrect grade, please discuss with me. If we cannot resolve the issue, you may talk to our departmental director of undergraduate studies or appeal the grade through a formal university process based on arithmetic/clerical error, arbitrariness, discrimination, harassment, or personal malice. To learn more, go to the Academic Advising Program website.

Counseling and psychological services

CAPS is strongly committed to addressing the mental health needs of a diverse student body through timely access to consultation and connection to clinically appropriate services, whether for short or long-term needs. Go to their website: <https://caps.unc.edu/> or visit their facilities on the third floor of the Campus Health Services building for a walk-in evaluation to learn more.

Title X resources

Any student who is impacted by discrimination, harassment, interpersonal (relationship) violence, sexual violence, sexual exploitation, or stalking is encouraged to seek resources on campus or in the community. Reports can be made online to the EOC at <https://eoc.unc.edu/report-an-incident/>. Please contact the University's Title IX Coordinator (Elizabeth Hall, interim – titleixcoordinator@unc.edu), Report and Response Coordinators in the Equal Opportunity and Compliance Office (reportandresponse@unc.edu), Counseling and Psychological Services (confidential), or the Gender Violence Services Coordinators (gvsc@unc.edu; confidential) to discuss your specific needs. Additional resources are available at safe.unc.edu.

Policy on non-discrimination

The University is committed to providing an inclusive and welcoming environment for all members of our community and to ensuring that educational and employment decisions are based on individuals' abilities and qualifications. Consistent with this principle and applicable laws, the University's Policy Statement on Non-Discrimination offers access to its educational programs and activities as well as employment terms and conditions without respect to race, color, gender, national origin, age, religion, creed, genetic information, disability, veteran's status, sexual orientation, gender identity or gender expression. Such a policy ensures that only relevant factors are considered and that equitable and consistent standards of conduct and performance are applied.

If you are experiencing harassment or discrimination, you can seek assistance and file a report through the Report and Response Coordinators (see contact info at safe.unc.edu) or the Equal Opportunity and Compliance Office, or online to the EOC at <https://eoc.unc.edu/report-an-incident/>.

Diversity statement

I value the perspectives of individuals from all backgrounds reflecting the diversity of our students. I broadly define diversity to include race, gender identity, national origin, ethnicity, religion, social class, age, sexual orientation, political background, and physical and learning ability. I strive to make this classroom an inclusive space for all students. Please let me know if there is anything I can do to improve, I appreciate suggestions.

Undergraduate testing center

The College of Arts and Sciences provides a secure, proctored environment in which exams can be taken. The center works with instructors to proctor exams for their undergraduate students who are not registered with ARS and who do not need testing accommodations as provided by ARS. In other words, the Center provides a proctored testing environment for students who are unable to take an exam at the normally scheduled time (with pre-arrangement by your instructor). For more information, visit <http://testingcenter.web.unc.edu/>.

Learning center

The UNC Learning Center is a great resource both for students who are struggling in their courses and for those who want to be proactive and develop sound study practices to prevent falling behind. They offer individual consultations, peer tutoring, academic coaching, test prep programming, study skills workshops, and peer study groups. If you think you might benefit from their services, please visit them in SASB North or visit their website to set up an appointment: <http://learningcenter.unc.edu>.

Writing center

The Writing Center is located in the Student and Academic Services Building and offers personalized writing consultations as well as a variety of other resources. This could be a wonderful resource to help with your writing assignments in this course (and any assignments in your other courses). You do not need a complete draft of your assignment to visit; they can help you at any stage! You can chat with someone in the writing center or set up an appointment on their website: <http://writingcenter.unc.edu>.